Unit 4, Lesson 5: Exploring Relationships in Circles

Benchmark

MA.7.GR.1.3 Explore the proportional relationship between circumferences and diameters of circles. Apply a formula for the circumference of a circle to solve mathematical and real-world problems.

Learning Target

☐ I can explain the relationship between the circumference and diameter of a circle.

4.5.1 Warm-Up: Ostrich vs Hummingbird

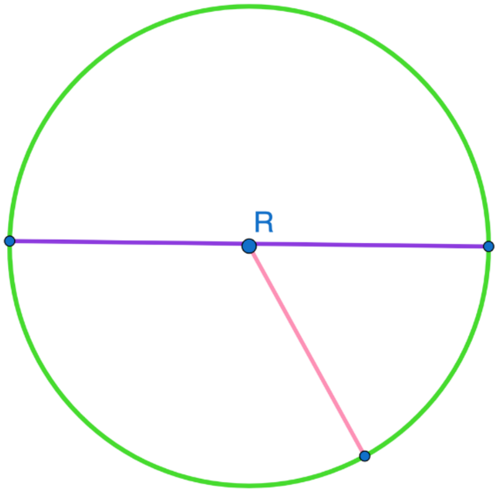
The average weight of a hummingbird is 0.15 ounces. The average weight of an ostrich is 230 pounds.

1. What is the ratio of the average weight of an ostrich to the average weight of a hummingbird?
   1 pound = 16 ounces.

4.5.2 Guided Instruction: Identifying Parts of a Circle

A ***circle*** is a perfectly round two-dimensional figure, where all points on the circle are equidistant from the center.

A circle with center R is shown.



The purple line segment is a diameter of the circle. The pink line segment is the radius of the circle. The center of a circle is the point that is the same distance from all points on the circle.

The plural of radius is radii. Circles have radii and diameters, both of which pass through the center of the circle.

The green part of the circle is the circumference.

1. Label each part of circle R by writing the name of each part.

A ***radius*** is a line segment extending from the center of a circle to a point on the circle.

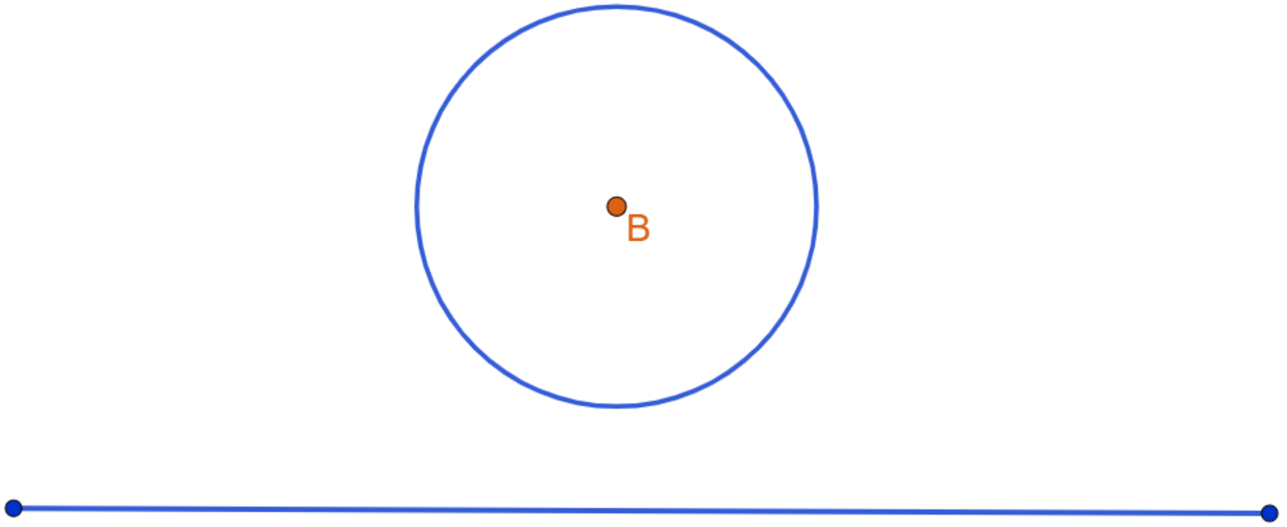
A ***diameter*** is a line segment from any point on the circle passing through the center to another point on the circle.

The length of a diameter is equal to the length of 2 radii.

The ***circumference*** is the distance around a circle.

If a circle were made of a piece of string and we cut the string and straightened it out, the circumference would be the length of the string.

Circle B is shown.



The line segment is approximately the same length as the circumference of circle B.

4.5.3 Exploration: Finding Circumference

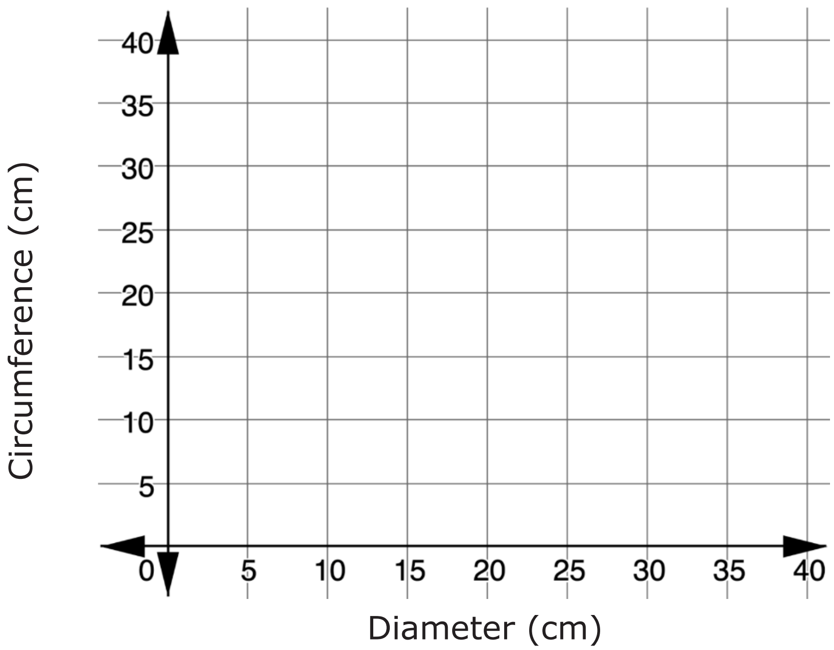
1. In the table, record the name of the five different circular objects your group is using.

2. Using a ruler, measure and record the diameter of each circle in centimeters (cm).

3. Using string, wrap the circular object with string using your finger to mark the circumference of the circle. Straighten out the string and use it to measure the length of the circumference in centimeters (cm) with a ruler. Then, record it in the table.

Object	Diameter	Circumference

A coordinate plane is shown with the horizontal axis representing the diameter and the vertical axis representing the circumference.



4. Plot the values for each object on the graph.

5. Discuss with your partner what you notice about the graph.

6. Using circumference (C) and diameter (D), find the value of the expressions listed in the table for each of the circular objects measured.

Object	$\boldsymbol{C + D}$	$\boldsymbol{C - D}$	$\boldsymbol{C \times D}$	$\boldsymbol{C \div D}$

7. What do you notice about the table?

8. Use the information in the table to complete the statement.

The value of

- ○ $C + D$
- ○ $C - D$
- ○ $C \times D$
- ○ $C \div D$

is approximately equivalent for each circular object. It is approximately ________.

9. Use the relationship you observed to predict the circumference of each circle using the given diameter. Record your predictions in the middle column.

Given Diameter	Predicted Circumference	Measured Circumference
1 foot (ft)		
6.25 inches (in.)		
$8\frac{1}{2}$ centimeters (cm)		

10. Use a string and a ruler to measure the circumference of each of the circles using the same units from the given diameter. Then, record the measured circumference of each circle in the table.

11. Describe the accuracy of your predictions.

4.5.4 Wrap-Up: Cool Down

1. Write an explanation to a classmate who might have missed today's lesson, in your own words, about the relationship between the circumference of a circle and its diameter.

